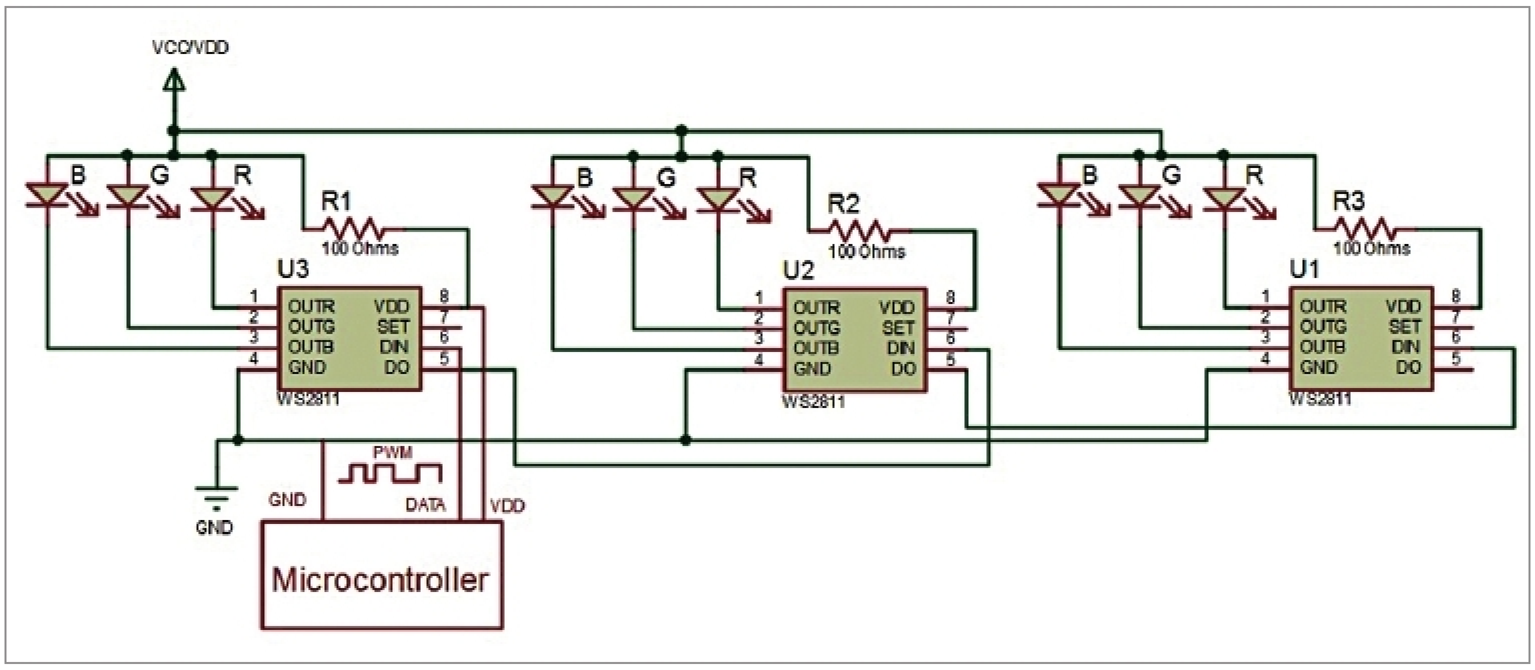

Fig. 3: Multiple NeoPixel LEDs connected together

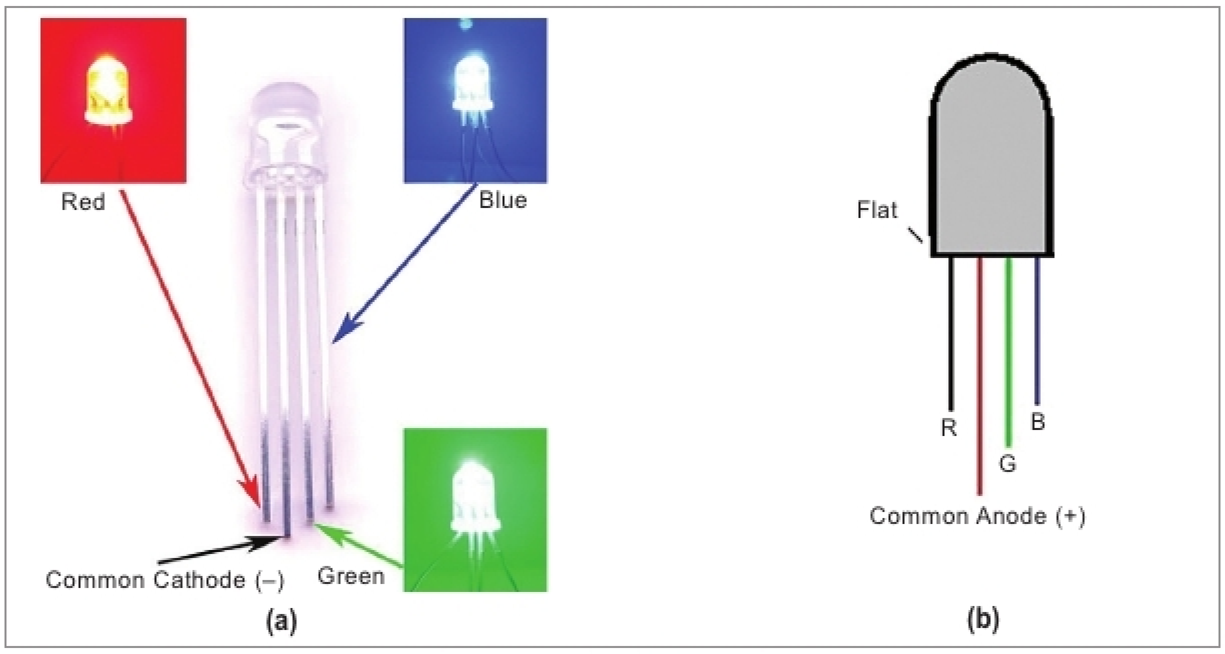

Fig. 4: RGB LED's pin diagrams: (a) common-cathode type, (b) common-anode type

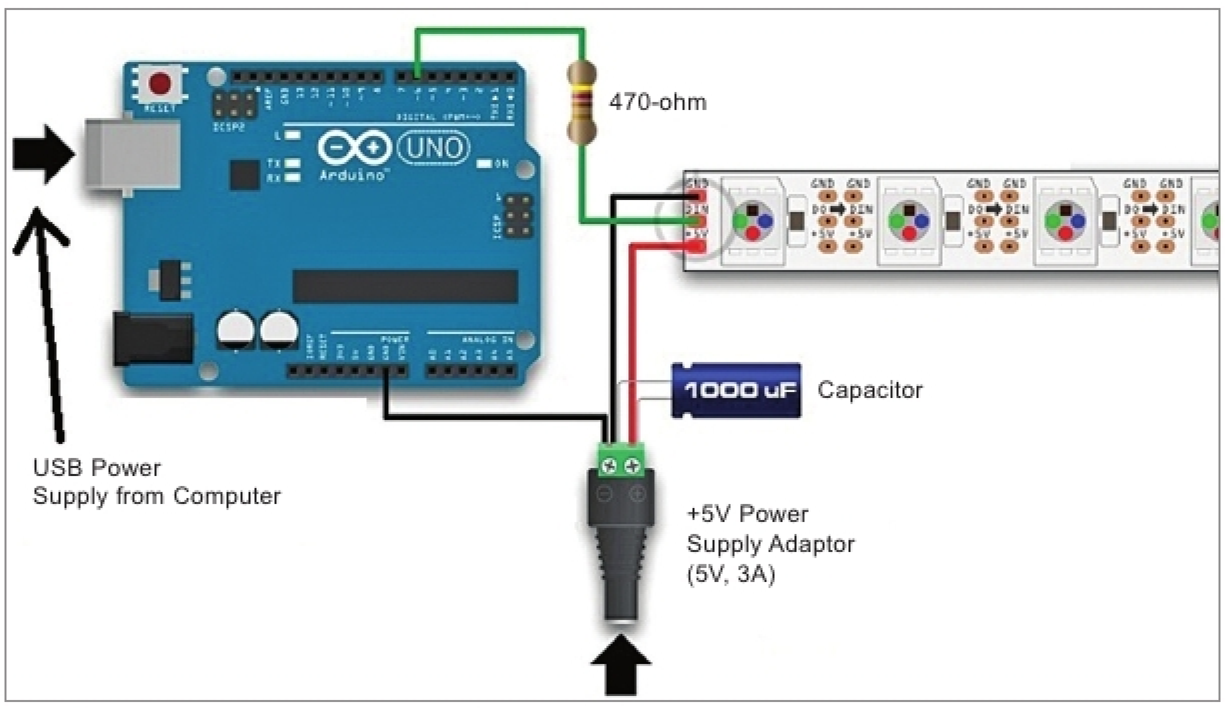

Fig. 5: Connection diagram of NeoPixel addressable LEDs using Arduino

Raspberry Pi, LaunchPad, and other popular platforms. Even Assembly language driver routines are available for WS2811-based products for some microcontrollers (MCU), including Atmel's popular 8-bit ICs, TI's TMS430 series and Microchip's PIC16F15xx devices.

Here, pulse width modulation (PWM) is used to control the NeoPixel LEDs. After resetting the controller of the LED chain by holding the input low for 50µs, the MCU program begins sending a stream of 24-bit colour codes (8-bit each for the R, G, and B LEDs) down the line. The first set of 24-bit RGB value is latched and displayed by the first LED pixel, which then allows the remaining bit stream to pass down the chain. The second LED pixel grabs RGB bit sequence, displays it, and so on, until the end of the chain is reached. The PWM pulses are varied to produce multiple colours with the combination of RGB LEDs.

Circuit and working

Connection diagram for the NeoPixel addressable LEDs using Arduino is shown in Fig. 5. Connect a 5V DC power supply to Arduino and the NeoPixel LED strip. Connect the +5V pin of the LED strip to the positive terminal of power supply. Do not connect 5V pin of the Arduino to the LED strip. Connect DIN and GND pins of the LED strip to digital pin 6 and GND pin of the Arduino, respectively. Make sure the GND pins of LED strip, Arduino, and 5V DC power supply are connected to each other.

This project uses a NeoPixel compatible 50-RGB LED strip and a power supply of 5V, 5A from a DC adaptor. This is because each NeoPixel consists of three LEDs (red, green, and blue), and each LED draws a maximum current of 20mA. That means, 60mA (20mA x 3) current is required for each NeoPixel LED, as shown in Fig.3. Total current requirement for fifty LEDs would thus be 50 × 60mA = 3000mA, or three amperes. So, the DC adaptor should have at least 3A or more current rating.

While using a DC power supply, or a large battery source, a 1000µF, 16V or higher capacitor should be added across the plus (+) and minus (-) terminals. This prevents the initial